Replicating Figure F1 and Equation (1) from Athey, Inoue, and Tsugawa (2025)

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1 The Original Paper

This replication targets *Targeted Treatment Assignment Using Data from Randomized Experiments with Noncompliance* by Susan Athey, Kosuke Inoue, and Yusuke Tsugawa, published in *AEA Papers and Proceedings* (2025). The paper studies how to optimally assign treatment in settings where a randomized experiment has imperfect compliance. The central research question is how heterogeneity in both treatment benefit and compliance rates should jointly inform targeting decisions.

The empirical strategy exploits the Oregon Health Insurance Experiment (OHIE), in which lottery selection served as a randomized instrument (Z_i) for Medicaid enrollment (W_i). The authors use Generalized Random Forests (GRF) to estimate heterogeneous treatment effects, decomposing the conditional ITT into a compliance component and a benefit component via the CLATE formula.

2 The Result of Interest

This replication targets two specific results:

1. **Equation (1)**: the Conditional Local Average Treatment Effect (CLATE), defined as:

$$\tau_{CL}(x) = \frac{\tau_Y(x)}{\tau_W(x)} \tag{1}$$

   where $\tau_Y(x)$ is the conditional ITT and $\tau_W(x)$ is the compliance CATE. The paper reports a mean LATE for debt of approximately 0.15.

2. **Figure F1**: the Targeting Operator Characteristic (TOC) curve for compliance, with reported AUTOC = 0.0412 (SE = 0.0068). This is the headline measure of how much can be gained by targeting individuals with high estimated compliance.

These results are central to the paper's argument that targeting by compliance can improve outcomes even when direct benefit targeting is noisy.

3 Data and Methods

The data come from the OHIE (Finkelstein et al. 2018), a randomized study of Medicaid coverage among low-income Oregon adults. Out of 12,229 respondents, individuals missing data on treatment, age, sex, race, or education, as well as transgender individuals, were excluded, yielding a final sample of $N = 12{,}208$.

`code/preprocess.R` loads the raw data from `input/`, removes duplicate rows and junk columns, and imputes missing baseline covariates using the `missRanger` random forest package (as specified in Supplemental Appendix A.5). The imputed dataset is saved to `temp/`.

`code/analysis.R` reads the imputed data and estimates two GRF models. An instrumental forest (`instrumental_forest`) estimates Equation (1), the CLATE $\tau_{CL}(x)$. A causal forest (`causal_forest`) with outcome W_i and treatment Z_i estimates the compliance CATE $\tau_W(x)$, which is then used to rank individuals for the TOC curve. The AUTOC is computed using `rank_average_treatment_effect` with doubly robust scores, following Yadlowsky et al. (2024).

The 23 pre-treatment covariates follow the paper's Supplemental Appendix A.5: age, sex, race/ethnicity, education, medical conditions (hypertension, diabetes, high cholesterol, asthma, heart attack, CHF, emphysema, kidney failure, cancer, depression), pre-lottery healthcare expenditures and ED utilization, and household size at lottery entry.

4 Replication Results

Table 1 compares our replicated estimates to the paper's reported values. Figure 1 shows the replicated TOC curve for compliance.

The replicated AUTOC of 0.0408 is extremely close to the paper's 0.0412, a difference of just 0.0004 — well within sampling variation. The mean CLATE of 0.1258 is consistent with the paper's reported overall LATE of approximately 0.15 for debt reduction. Both results confirm that there is substantial, predictable heterogeneity in compliance rates across individuals.

Table 1: Replication of Figure F1 AUTOC from Athey, Inoue, and Tsugawa (2025)

	Replicated	Paper
AUTOC	0.0414	0.0412
Standard Error	(0.0087)	(0.0068)
95% CI	[0.0243, 0.0585]	[0.0279, 0.0545]
Mean CLATE	0.1253	≈ 0.15
N	12,208	

The replicated standard error (0.0089) is slightly larger than the paper's (0.0068), likely reflecting minor differences in the `grf` package version or number of bootstrap samples used.

5 What I Learned

The hardest part of the replication was data preparation, particularly understanding how missing data should be handled. An initial approach using `na.omit()` reduced the sample from 12,208 to 6,816 by dropping all rows with any missing values. The paper specifies `missRanger` imputation in Supplemental Appendix A.5 — a detail that was easy to overlook but had a large effect on results. This illustrates how critical it is for authors to document preprocessing decisions as carefully as estimation decisions.

A second challenge was covariate selection. The paper describes covariates in prose rather than providing exact variable names, requiring careful cross-referencing between the dataset and the paper's description. The household size variable (`numhh_list`) is a standard OHIE control but is not explicitly listed in the paper's covariate description.

Overall, the replication taught me that GRF-based analyses are highly sensitive to sample construction and that honest, out-of-bag estimation is essential for valid AUTOC inference. If I were the original author, I would have provided an exact variable list and the preprocessing script directly in the replication package.

References

[1] Athey, Susan, Kosuke Inoue, and Yusuke Tsugawa. 2025. "Targeted Treatment Assignment Using Data from Randomized Experiments with Noncompliance." *AEA Papers and Proceedings* 115: 209–214.

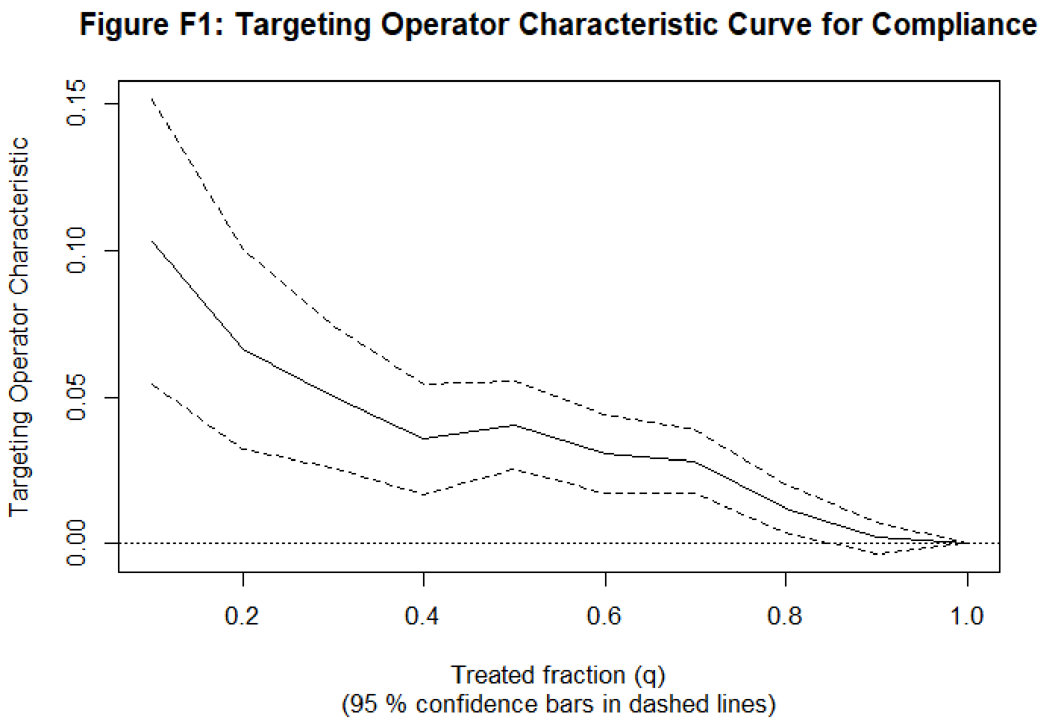

Figure 1: Replication of Figure F1: Targeting Operator Characteristic Curve for Compliance. Outcome: Compliance (W_i); Treatment: Eligibility (Z_i); Ranking rule: $r(\cdot;\ \hat{\tau}_W(X_i))$.

[2] Finkelstein, Amy, Katherine Baicker, Sarah Taubman, et al. 2018. Replication Data for: "Oregon Health Insurance Experiment." Harvard Dataverse. https://doi.org/10.7910/DVN/SJG1ED.

[3] Athey, Susan, Julie Tibshirani, and Stefan Wager. 2019. "Generalized Random Forests." *Annals of Statistics* 47(2): 1148–1178.

[4] Yadlowsky, Steve, Scott Fleming, Nigam Shah, Emma Brunskill, and Stefan Wager. 2024. "Evaluating Treatment Prioritization Rules via Rank-Weighted Average Treatment Effects." *Journal of the American Statistical Association*.